What Did Ten Points of Mean Recall Buy? Separating Group-Level from Case-Level Gains in Scene Graph Generation

Anonymous CVPR submission

Paper ID *****

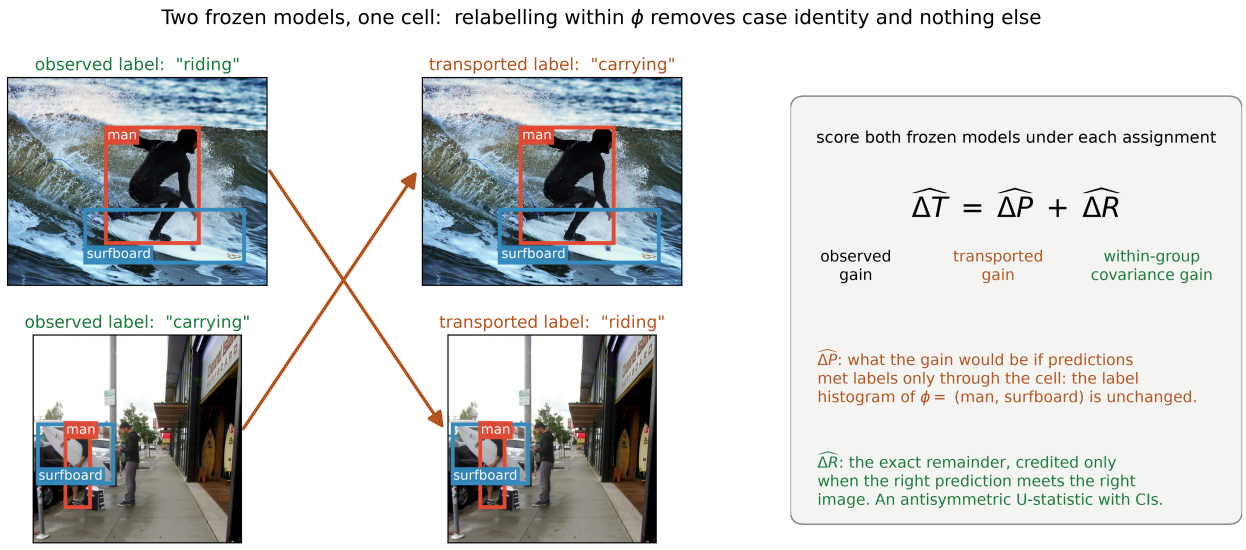


Figure 1. **Within-group relabelling.** Two Visual Genome relations share the subject–object pair (man, surfboard). Swapping their labels keeps the pair's predicate histogram exactly as it was but breaks the link between each image and its own label. Scoring two frozen models under the observed and the swapped labels splits their score gain exactly: $\widehat{\Delta P}$ is the part that survives the swap — earned at the level of the group — and $\widehat{\Delta R}$ is the covariance between each relation's prediction and its own label, which better ranking within the group can supply, as can a moved operating point.

Abstract

Debiasing methods for scene graph generation are judged by mean recall, and one of the best known, Total Direct Effect (TDE), raises it on Visual Genome from 14.6 to 24.8. What did those ten points buy? A model raises mean recall either by predicting rare predicates more often within each subject–object pair, or by telling which individual relations carry them, and the metric credits both alike. Scoring each prediction against the label of another relation from the same pair keeps the pair's predicate frequencies but breaks the link between each relation and its own label. This splits the difference between two frozen models, in a proper score or in mean recall itself, exactly into a group-level part and a case-level covariance, in $O(NK)$ time with image-clustered intervals. We audit three released Neural-Motifs checkpoints in PredCls. For TDE, 87% of the mean-recall@50 gain (graph-constrained, micro-pooled; interval 83–91%) is group-level when cells are subject–object class pairs, and a logit adjustment that changes no ranking within any pair recovers $+0.094$ of its $+0.097$ at under a tenth of TDE's cost in recall. The within-pair AUC excludes a change in TDE's ranking larger than 0.0034 with cells weighted as in the split, but only one larger than 0.0077 when every comparison counts once, and TDE's case-level part grows when the label mix is made uniform within pairs. DLFE, against its own released baseline, gains 11 points, 94% group-level, and its within-pair ranking does not move

detectably; IETrans, against a separately trained baseline, gains 21 points, 97% group-level, and its within-pair ranking falls. SGCls and SGDet replications of TDE agree.

1. Introduction

Scene graph generation is evaluated on a benchmark whose labels are unusually predictable from something every model sees. Given the ordered object pair (man, surfboard), the predicate on Visual Genome [20] is very often riding, and a lookup table of training frequencies is famously competitive [50]. Recall@K rewards models that echo this prior, so the field now leads with mean recall, which averages recall over predicates and so rewards getting the rare ones [5, 40]. Debiasing methods are now judged by how far they raise it. Total Direct Effect (TDE) [41], among the most cited of them, raises mean recall@50 of a MOTIFS model from 14.6 to 24.8 on the canonical VG150 split, in our evaluation of the released checkpoint.

What did those ten points buy? A model can move probability towards rare predicates *within each subject–object pair* — say carrying more often for (man, surfboard) — without becoming any better at deciding *which* relations are a man carrying rather than riding. Or it can learn to make that decision. The metric credits both alike. Existing remedies — head/body/tail-stratified recall [23], independent mean recall [25], zero-shot recall [27], rarity-stratified accuracy [19] — slice one model's performance differently; none attributes the gain *between* two models to one route or the other.

An exact split. Take two frozen models and a test set grouped by a discrete feature both observe, here the subject–object pair (Fig. 1). Score each model's prediction for one relation against the label of a *different* relation in the same group. This keeps each group's predicate frequencies, so gain earned at the level of the group survives it, and breaks the link between a relation and its own label, so gain earned on individual cases does not. The observed gain splits exactly into a *group-level* part $\widehat{\Delta P}$ and a *case-level* remainder $\widehat{\Delta R}$, a within-group covariance between the change in the predictions and the label (Theorem 1), for any proper score and for mean recall itself, with no fitted reference model, in $O(NK)$ on cached predictions, with image-clustered intervals.

What it finds. We re-ran TDE's released checkpoint, replayed the official evaluator exactly so that recall can be computed under any labelling, and split mean recall itself (Sec. 5). Of the ten points, 87% is group-level (of the micro-pooled recall, $+0.0972$; the official statistic moves $+0.1017$); the fifteen rarest predicates contribute almost nothing. A thresholded metric also earns case-level credit when its operating point moves, so we read the remainder against a control: subtracting the training predicate prior from the baseline's logits [29], a correction known to rival debiasing [6, 43] that changes no ranking within any group, recovers $+0.0941$ of the $+0.0972$ with the same split at under a tenth of TDE's cost in R@50, and against it TDE's case-level part is -0.0008, straddling zero (at $K = 50$; the control's match is specific to that cut-off). The within-group AUC excludes a change in TDE's ranking larger than 0.0034 under the split's weighting and larger than 0.0077 when every comparison counts once. A proper score's case-level part is no ranking measure: TDE's is $+0.0045$ under the pre-declared matching protocol, but removing the frequency prior, which ranks every pair as before, gives $+0.0082$, and the part changes sign with the protocol and the label mix. DLFE [6], released with its own baseline, gains 11 points, 94% group-level, with no detected change in within-pair ranking; IETrans [51], which relabels training data within object pairs, gains 21 points against a separately trained baseline, 97% group-level, and its relation-weighted AUC falls. The converse appears in a controlled illustration on real pixels, not an SGG system (Sec. 6): a predictor reading frozen CLIP [32] crops, ranked last by every aggregate, carries $+0.0033$ more case-level covariance than a geometry model, yet ranks worse under the relation-weighted AUC (better, not significantly, by comparison).

Contributions. (i) An exact, model-free decomposition of the difference between two frozen models into a group-level part and a within-group covariance, the paired-gain form of Yates's covariance decomposition [47], for every Bregman score and for label-weighted hit rates such as mean recall, with cluster-robust inference, an operating-point control and a threshold-free AUC. (ii) Validation that the case-level part is zero when it should be, covers at the nominal rate under image clustering, and separates calibration, prior and shortcut effects, with protocols for the confounds it does not. (iii) Audits of three released debiasing checkpoints that split their mean-recall gains exactly, with the mechanism behind TDE's.

What is not claimed. The scene-graph audits are PredCls, where every model sees the cell, except a replication of TDE in SGCls and SGDet (Sec. 5), where the two models share every predicted object. The split is conditional on the declared grouping: $\widehat{\Delta R}$ measures case-level discrimination *among relations sharing an object pair*, and credits any signal varying inside a pair, including a shortcut nobody recorded. It is not invariant to recalibration — rescaling a model's confidence moves score between the two parts while changing no ranking — which is why the comparisons whose models differ visibly in confidence, TDE's and

the CLIP model's, are read confidence-matched. For mean recall the split is of the micro-averaged version, which pools relations across images; it tracks the official per-image average but is not it. And a proper-score case-level part near zero can depend on the calibration protocol and on the label mix; Appendices O and Q report every configuration. These limits are taken up in Sec. 7.

2. Related work

Bias and its evaluation in scene graph generation. Scene graph generation was posed as structured prediction over a whole image, solved by message passing between object and relation nodes [45]; MOTIFS then showed a frequency predictor of the predicate given the object pair to be surprisingly competitive [50]. Mean recall became the headline metric, and debiasing methods now span the causal TDE intervention [41], class-balanced sampling and group-wise learning [11, 13], predicate-rebalancing message passing [23], predicate-correlation and hierarchical losses [28, 46, 48], prototype embeddings [52], and correction or relabelling of the training labels [22, 51]. Correction by the predicate prior is known to rival them: post hoc, DLFE [6] recovers unbiased predicate probabilities from estimated label frequencies and Structured Sparse R-CNN [43] applies logit adjustment [29]; in training, RTPB [4] resists a prior bias. The metrics themselves have been shown to reward pathologies [25], and open-vocabulary methods inherit them [24]. These works compare methods by their metrics; we ask which part of a metric's change is group-level. Benchmark audits outside scene graphs find the same pattern: VQA models collapse when answer priors change [1], multimodal models exploit non-visual shortcuts [15, 53], and gains are credited to mechanisms that did not produce them [37, 42]. Long-tailed recognition faces the same question for re-weighting and logit adjustment [3, 29]. All of these characterise a *single* model; we account for the gain *between* two.

Proper-score decompositions. Conditional on a grouping, a proper score splits into reliability and resolution terms [9, 30], for general proper scores [2]. Our case-level term is not that resolution term: it descends from the *covariance* decomposition of the Brier score due to Yates [47], which a monotone recalibration moves while the classical resolution is invariant to it. The two coincide only when both models are calibrated within every group. Variance estimators exist for single-model decompositions [38], not for a contrast between two models. The grouping loss [31] is the single-model relative of our case-level part: what calibration leaves behind because inputs sharing a confidence level differ; ours contrasts two models within groups declared in advance. Transporting labels within a group is a within-group permutation of the input–label pairing, so one model's case-level part is a conditional permutation importance [39] or model reliance [14] of the non-grouping part of its input, and the case-level part of a gain is the difference of the two models' (Proposition 2). What the split adds is the exact additive accounting of a paired gain, including thresholded metrics, and paired inference for it.

Comparing two predictors. Inference on a paired score difference is the Diebold–Mariano test [12, 16]; at a single group our total-gain interval is its clustered form (Corollary 3), so the decomposition adds a split to what it reports. DeLong *et al*. [10] compare correlated ROC areas with a paired U-statistic, the device we use for a group-conditional gain; the case-level term is an order-two U-statistic [18] with a Hájek projection [36] as its influence function.

3. Splitting a score gain

Setup. Let $q_0(\cdot \mid x)$ and $q_1(\cdot \mid x)$ be the predictive distributions of an old and a new frozen model over K labels, and $S(q, y)$ a bounded proper score, larger being better. Our default is the quadratic score $S(q, y) = 2q_y - \|q\|_2^2$, negative Brier loss up to a label-only constant; the log score is the second case we report. Let $C = \phi(X)$ be a discrete grouping both models observe, declared before the audit — for relation prediction, the ordered subject–object class pair. We call a value of C a *cell*. The gain vector

$$H_x(y) = S(q_1(\cdot \mid x), y) - S(q_0(\cdot \mid x), y), \qquad y = 1, \dots, K, \tag{1}$$

can be evaluated at a counterfactual label without rerunning either model, and that is all the construction needs.

Transport. For a cell c, let $\Delta T_c = \mathbb{E}[H_X(Y) \mid C = c]$ be the observed gain and $\Delta P_c = \mathbb{E}_{H|c}\mathbb{E}_{Y'|c}[H_X(Y')]$ the gain when each case is scored against an *independent* label Y' drawn from the same cell, and set $\Delta R_c = \Delta T_c - \Delta P_c$. Transport keeps the cell and its label distribution and removes only the pairing between a case and its own label.

Theorem 1 (Transport and covariance identities)**.** *For any integrable score contrast H,*

$$\Delta T_c = \Delta P_c + \Delta R_c,$$

$$\Delta R_c = \sum_{y=1}^{K} \mathrm{Cov}(H_X(y), \mathbb{1}\{Y = y\} \mid C = c)\,. \tag{2}$$

For the quadratic score, with $\Delta q_y(X) = q_1(y \mid X) - q_0(y \mid X)$,

$$\Delta R_c = 2 \sum_{y=1}^{K} \mathrm{Cov}(\Delta q_y(X), \mathbb{1}\{Y = y\} \mid C = c)\,. \tag{3}$$

Consequently, if the gain vector is a function of c alone, $\Delta R_c = 0$.

Eq. (3) is what the split means. The new model earns case-level credit only insofar as its probability *change* rises on the label the individual case actually carries, relative to other cases in the same cell. Anything constant within a cell — shifting probability towards rare predicates for every (man, surfboard) relation alike — lands in ΔP, which is why we call it the group-level part. It is *not* simply "prior fit": under the quadratic score ΔP_c is the improvement in how close the model's mean forecast sits to the cell's label frequencies *minus* the increase in the dispersion of its forecasts (Proposition 1). The covariance form is due to Yates [47] and generalises: for every Bregman score, including the log score, ΔR_c is the same covariance with Δq_y replaced by the contrast of the y-th gradient coordinate (Theorem 2). All proofs are in the supplementary.

Why it matters which part a gain is in. In one cell with three cases labelled 1 and one labelled 2, against an old model predicting $(\frac{1}{2}, \frac{1}{2})$, a model predicting the cell's frequencies $(\frac{3}{4}, \frac{1}{4})$ for every case gains $+\frac{1}{8}$ of quadratic score, all group-level; a model equally far from the old one but pointed at each case's own label gains $+\frac{3}{8}$, all case-level. Shift the cell's labels to $(\frac{1}{4}, \frac{3}{4})$ and the first falls to $-\frac{3}{8}$, worse than the model it replaced, while the second stays at $+\frac{3}{8}$. In general, with $m_y(z) = \mathbb{E}[H_X(z) \mid Y = y, C = c]$, $\Delta R_c = \sum_{y<z} p(y)p(z)\, D(y,z)$ where $D(y,z) = m_y(y) - m_y(z) + m_z(z) - m_z(y)$: a label shift fixes every D and moves only the weights, so a case-level gain is less exposed to a shift than a group-level one, though with three or more labels it can still reverse (Appendix C).

Estimator. In a cell with $n_c \geq 2$ cases, write $H_i(y) = H_{x_i}(y)$ and estimate

$$\widehat{\Delta T}_c = \tfrac{1}{n_c} \textstyle\sum_i H_i(y_i), \qquad \widehat{\Delta P}_c = \tfrac{1}{n_c(n_c-1)} \textstyle\sum_{i \neq j} H_i(y_j), \tag{4}$$

$$\widehat{\Delta R}_c = \tbinom{n_c}{2}^{-1} \textstyle\sum_{i<j} A_{ij}, \tag{5}$$

$$A_{ij} = \tfrac{1}{2}\{H_i(y_i) + H_j(y_j) - H_i(y_j) - H_j(y_i)\}. \tag{6}$$

These satisfy $\widehat{\Delta T}_c = \widehat{\Delta P}_c + \widehat{\Delta R}_c$ exactly in every sample, and $\widehat{\Delta R}_c$ is an unbiased order-two U-statistic [18] for ΔR_c (Theorem 3). Transporting only to the *other* cases' labels matters: the all-case plug-in shrinks the case-level gain by $(n_c - 1)/n_c$ (Proposition 5). With m_{cy} the count of label y in cell c, case i's transported score is

$$P_i = \frac{\sum_y m_{cy} H_i(y) - H_i(y_i)}{n_c - 1}, \tag{7}$$

so the whole estimator costs $O(NK)$ on cached predictions, with no fitted model, held-out fold or smoothing parameter. Cells are weighted by size; singletons cannot identify a within-cell counterfactual and are excluded and reported.

Inference. The Hájek influence contributions of the U-statistic (Appendix F), summed within each image, give one-way cluster-robust intervals, since relations from one photograph are not independent; the estimator is asymptotically normal under bounded scores and non-dominating clusters (Theorem 4), and at a single cell the total-gain interval is a clustered Diebold–Mariano test [12] (Corollary 3).

Confidence matching. Because ΔR is a covariance between probability *movements* and labels, rescaling a model's confidence moves score between the two parts while changing no ranking (Sec. 4), so models that differ in calibration are not compared on raw outputs. We split the images at random, fit each model a recalibration by maximum likelihood on one half, and split the gain on the other. The family is fixed before any gain is read, by one criterion: it must be rich enough that a model differing from another by a shift of each label's log-odds that is the same in every cell, which changes no ranking, has no case-level part under it. A temperature alone fails this, a temperature with a per-class bias meets it exactly and still preserves every within-cell ranking, so the latter is the primary protocol and the temperature-only split is reported beside it.

Splitting mean recall. Nothing in Theorem 1 or the estimator uses properness: both hold for any contrast $H_x(y)$ that can be evaluated at a counterfactual label. Mean recall@k is one. Let $h_x^{(m)}(y) \in \{0, 1\}$ say whether model m would recall relation x among its top k if its label were y; replaying the benchmark's matching rules on cached scores gives it for every y (Appendix O). Pooling relations, mean recall is $\sum_i w(y_i) h_{x_i}(y_i)$ with $w(y) = 1/(Kn_y)$, so $H_x(y) = Nw(y)[h_x^{(1)}(y) - h_x^{(0)}(y)]$ makes $\widehat{\Delta T}$ its change; transport preserves every n_y, and restricting w to one tier or predicate splits that share exactly. One difference matters for reading the result. A hit is a thresholded decision, so moving a model's operating point — subtracting a log-prior from its logits, say — flips first the cases it already scored as most like the favoured label, and those carry that label more often: a case-level part with no new discrimination. We therefore read the split against an *operating-point control*, the old model's logits shifted by the training log-prior [29], and pair it with a threshold-free test, the *within-cell AUC*: the probability that a case labelled y has larger log-odds $\log q(y) - \log q(z)$ than one labelled z in the same cell. Averaged with weight $n_c(y)n_c(z)$, the number of comparisons, it weights a cell by about n_c^2; with weight $n_c(y)n_c(z)/n_c$ it combines cells as the split does; we take the latter as primary and report both. A temperature, and any change constant within a cell, leaves either unchanged.

Choosing the grouping. The split is conditional on ϕ. Merging cells changes it by an explicit between-cell covariance term of either sign (Proposition 3), so the finest grouping the benchmark's construction defends is the conservative choice. A relevant variable left out of ϕ is the same term with the fully adjusted cell as the fine partition; Proposition 4 bounds it by one elicited correlation, after the robustness value of Cinelli and Hazlett [7].

4. Does the split measure what it claims?

A released checkpoint cannot validate the split, because what the model can see is unknown. We validate it where the answer is known, first in simulation and then on Visual Genome predictors whose information content we control. All intervals are 95%.

Simulation. With 24 cells over five labels, an old model equal to the cell prior and a new one interpolating towards an oracle, the estimated case-level gain grows linearly with the injected signal; under a within-cell null the randomization test rejects in 0.030 of 200 datasets at level 0.05. With relations clustered into images by a shared per-image effect and Zipf-distributed cell sizes, the image-clustered interval covers in 94.0% of 500 runs, against 88.6% for one that treats relations as independent; with VG150's own cell sizes it covers 96.3%, and so do the mean-recall and share intervals, and only when every cell holds two cases does it fall short, to 81.5% (Appendix J).

Three designs check that the two parts separate what they claim to. A *prior-only* improvement, where the new model corrects a noisy estimate of the cell prior, gives a case-level gain centred at zero (0.0002, sd 0.0035). An *unrecorded within-cell shortcut* — a binary feature varying inside each cell that the new model uses — is credited to the case-level part (+0.041, sd 0.004): the split measures all within-cell signal relative to the declared grouping, shortcut or not. And a *calibration-only* change, where the new model is a temperature-softened copy of an overconfident one carrying identical case-level information, is credited a large spurious case-level *loss* of −0.488 on raw outputs, which the confidence-matching protocol of Sec. 3 reduces to −0.0003 (sd 0.0017). A *class-wise shift of the operating point*, a fixed logit offset per label, changes no within-cell ranking yet survives temperature matching (+0.0106 for a logit-adjustment offset) and vanishes under a temperature with a per-class bias (0.0000, sd 0.0003; Appendix Q.4): the criterion of Sec. 3 at work.

Controlled predictors on VG150. We train predicate classifiers whose inputs we control on a VG150-style split reconstructed from the released annotations (same vocabulary, images divided 70/30 between fitting and audit; Appendix N.1), not the canonical split of Sec. 5: a frequency table on the class pair (FREQ), an MLP on class embeddings (MLP-CLASS), and MLPs adding twelve box-geometry features (MLP-SPATIAL, MLP-SPATIAL+). The audit covers 227,337 of 229,605 test relations (99.0%) in 6,346 class-pair cells.

MLP-CLASS beats FREQ by 0.03386 of quadratic score and the split assigns *all* of it to the group-level part: the case-level gain is 0.00000 with a zero-width interval, an algebraic zero (both models are functions of the class pair, so Theorem 1 forces it) and the sharpest check that nothing is credited to case-level evidence when none exists. Adding box geometry yields a case-level gain of 0.01439 (CI $[0.01373, 0.01504]$, $p = .002$) that no aggregate isolates; against MLP-CLASS the total gain of 0.00885 even *understates* it, because the group-level part falls by 0.00554.

5. What TDE's ten points bought

Tang *et al*. [41] release a causal MOTIFS PredCls checkpoint that yields two models from one set of weights: the baseline (MOTIFS-SUM), whose predicate logit sums a union-region visual term, a context term and a frequency prior indexed by the object pair, and TDE, which subtracts the same sum with an averaged context and so keeps only the context term minus its average. We ran it once with the authors' code on the canonical VG150 test split, saving every branch's logits, and replayed the official evaluator offline (Appendix O). The replay reproduces the official numbers to the fourth decimal: baseline $R@50 = 0.6612$ and $mR@50 = 0.1459$, TDE 0.4588 and 0.2476; without the graph constraint TDE's mean recall *falls*, from 0.3260 to 0.2981. The audit covers 183,642 relations over 26,446 images in 7,127 cells; 98.9% fall in the 5,044 cells holding at least two.

An operating-point control. We add one model that needs no training: the baseline's logits minus $\tau \log \pi(y)$, the training predicate prior, with $\tau = 1$ [29]. Within a cell this adds the same constant to every log-odds $\log q(y) - \log q(z)$, so it changes no within-cell ranking: whatever case-level part it shows is existing discrimination spent at a new operating point, not new discrimination. It reaches $mR@50 = 0.2343$ at $R@50 = 0.6437$.

Mean recall, split. Two quantities are primary: the graph-constrained $mR@50$ split and the relation-weighted within-cell AUC; every other protocol, score and weighting is secondary and collected in Tab. 27. Pooled over identified relations, mean recall@50 rises by 0.0972 (0.1017 on the official per-image average). Of that, +0.0841 is group-level and +0.0130 case-level, interval $[+0.0093, +0.0168]$ (Tab. 1): 87% of the gain, interval $[83\%, 91\%]$, survives

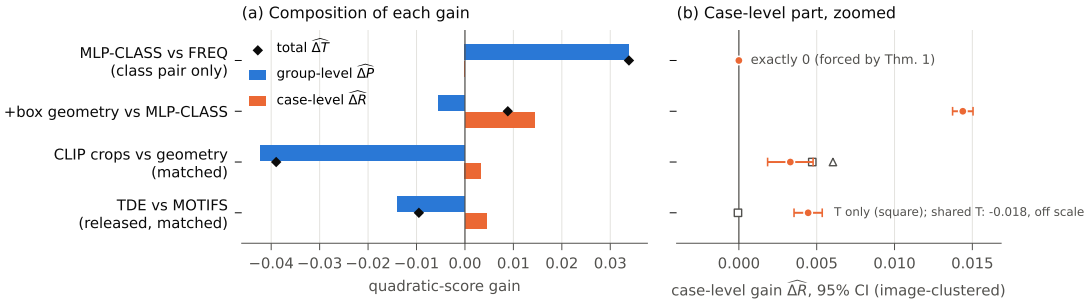


Figure 2. **Where each proper-score gain lives** (quadratic score). (a) Total gain $\widehat{\Delta T}$ (diamond) split exactly into group-level $\widehat{\Delta P}$ and case-level $\widehat{\Delta R}$. (b) $\widehat{\Delta R}$ with its image-clustered 95% interval. Class-pair-only models differ purely at group level (Theorem 1); geometry and CLIP crops buy case-level gain, CLIP despite losing overall; TDE's is group-level, with a small case-level residue. CLIP and TDE rows are matched by a temperature and a per-class bias per model (square: temperature only; triangle: one shared temperature).

Table 1. **Mean recall@50 split** (graph constraint, micro-pooled) against the MOTIFS baseline, except TDE−LA. LA: the logit-adjusted baseline; DLFE: against its own released baseline; IETrans: the released IETrans+Rwt checkpoint against a separately trained baseline. Case-level intervals image-clustered. ΔAUC: within-cell AUC change, weighted as the split weights cells (r, primary) or by comparison (c), image bootstrap; no calibration or group-level shift moves it (LA's residue is float32 rounding).

	ΔmR	group	case [CI]	ΔAUC_r	ΔAUC_c
TDE	+.0972	+.0841	+.0130	−.0015	+.0026
			[+.0093,+.0168]	[−.0036,+.0018]	[−.0028,+.0086]
LA	+.0941	+.0802	+.0139	+.0003	+.0001
			[+.0101,+.0176]	[+.0001,+.0006]	[−.0000,+.0001]
TDE−LA	+.0031	+.0039	−.0008	−.0018	+.0026
			[−.0050,+.0033]	[−.0039,+.0014]	[−.0029,+.0086]
DLFE	+.1112	+.1046	+.0065	+.0029	+.0018
			[+.0010,+.0120]	[−.0004,+.0056]	[−.0041,+.0064]
IETrans	+.2204	+.2143	+.0060	−.0235	−.0093
			[−.0031,+.0151]	[−.0275,−.0182]	[−.0181,+.0004]

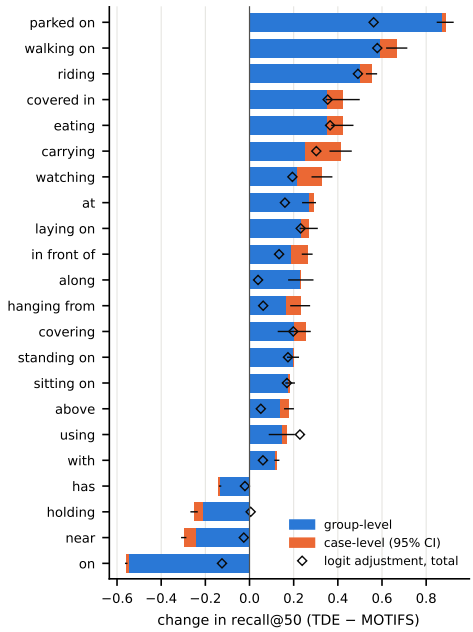


Figure 3. **Where TDE's mean recall went**, for every predicate whose recall@50 moved by at least 0.12: group-level part (blue), case-level part (orange) with its image-clustered 95% interval, and the total under logit adjustment of the baseline (diamond), which changes no within-cell ranking.

shuffling every label among the relations that share its object pair. With cells refined by the object's position or the subject's size the share is 90%, with the subject class alone 18% (Appendix K); transport across images only gives +0.0140, and the official per-image statistic, split by shuffling labels within cells, is 84% group-level (+0.0168, [+0.0111, +0.0226]; Appendix Q.1). It lives in the body of the distribution: by training frequency (9 head predicates above 10,000 instances, 26 body, 15 tail below 500), the body gains +0.1224, of which +0.1082 is group-level, the head loses 0.0255, and the tail gains 0.0003 (the baseline recalls no tail predicate; TDE recalls the predicate to at 0.016). Figure 3 shows the mechanism one predicate at a time. Recall of parked on rises from 0.000 to 0.886, almost all of it group-level: TDE now says parked on for (car, street). Recall of on falls by 0.558 in the same way.

The case-level part is matched by the control. The control reproduces the whole picture: +0.0941 of mean recall, +0.0802 group-level and +0.0139 case-level. Against it, TDE's case-level part is −0.0008, interval [−0.0050, +0.0033]. So a case-level part of a thresholded metric is not by itself evidence of new discrimination. When the boundary between on and parked on moves, the cases that cross it are those the baseline already scored as more parked on-like, and they are more often labelled so; the covariance is real, but bought with discrimination the

baseline had. The direct test is the within-cell AUC, which no prior shift or recalibration can move. Weighted as the split weights cells, the primary weighting, TDE's change is -0.0015, $[-0.0036, +0.0018]$; by comparison $+0.0026$, $[-0.0028, +0.0086]$. A null needs a bound: the 90% bootstrap intervals exclude a change larger than 0.0034 under the primary weighting ($[-0.0034, +0.0015]$) and larger than 0.0077 by comparison ($[-0.0022, +0.0077]$). For scale, box geometry buys 0.063 and 0.032 over the class pair (Sec. 6), so TDE's ranking matches the baseline's within a tenth of that under the primary weighting and within a quarter by comparison (Appendix Q.4). Against the control the mean-recall case-level part excludes only changes above 0.0043, a third of TDE's own, too wide to call it zero.

The proper score, matched. Under the primary protocol (Sec. 3), which makes the control identical to the baseline, TDE's quadratic-score change is -0.00951: -0.01396 group-level and $+0.00445$, interval $[+0.00354, +0.00536]$, case-level ($+0.02543$ in log score; Fig. 2; cross-fitted over the two halves $+0.00399$, $[+0.00334, +0.00463]$). That part is not a ranking measure: removing TDE's frequency prior, which leaves every within-cell ranking as it was (AUC change $+0.0001$), gives $+0.00817$, $[+0.00757, +0.00877]$, under the same protocol, since a per-class bias cannot absorb a shift that differs from pair to pair; ranking is read from the AUC alone. The protocol also decides the sign: with a temperature alone the part is -0.00006, $[-0.00120, +0.00109]$, and with one temperature shared by both models -0.0184; over 20 calibration halves the primary part stays positive, and merging synonymous predicates changes none of this (Appendices Q.1, Q.2 and Q.4).

What that licenses. On this checkpoint the ten points of mean recall are a change in which predicates the model favours within each object pair. A one-line prior shift that ranks the relations of every cell exactly as the baseline did reproduces the gain, its split and its tier profile at under a tenth of TDE's cost in $R@50$ (0.0175 against 0.2023), and a better $F@50$, the harmonic mean of $R@50$ and $mR@50$ [51] (0.344 against 0.322). That a prior correction rivals debiasing is known [6, 43]; the split shows which part of TDE's gain it reproduces. Table 2 gives all six recall protocols: the share runs from 77% at $K = 20$ to 86% at $K = 100$ under the graph constraint; without it TDE's mean recall falls, and its case-level part over the control turns positive at $K = 100$, $+0.0148$, $[+0.0064, +0.0232]$. The control's match of TDE's gain is specific to $K = 50$ under the constraint: TDE leads it by $+0.0300$ at $K = 100$ and trails it by 0.0234 at $K = 20$ and by 0.08–0.15 without the constraint. Group-level is not a lesser gain (mean recall rewards moving rare predicates within a pair by design), nor is the share forced by the metric: an oracle gains $+0.5068$ in quadratic score, all case-level ($+0.8041$ against -0.2973).

Table 2. **The split under all six recall protocols**, micro-pooled mean recall with (mR) and without (ng-mR) the graph constraint at $K = 20, 50, 100$; case-level intervals image-clustered.

	TDE vs baseline				TDE vs LA		
Protocol	ΔmR	group	case [CI]		ΔmR	case [CI]	
mR@20	+.0598	+.0462	+.0135	iny[+.0101,+.0170]	−.0234	+.0013	iny[−.0024,+.0
mR@50	+.0972	+.0841	+.0130	iny[+.0093,+.0168]	+.0031	−.0008	iny[−.0050,+.0
mR@100	+.1280	+.1099	+.0182	iny[+.0144,+.0220]	+.0300	+.0034	iny[−.0010,+.0
ng-mR@20	−.0037	−.0134	+.0096	iny[+.0058,+.0135]	−.0792	+.0013	iny[−.0035,+.0
ng-mR@50	−.0337	−.0464	+.0127	iny[+.0080,+.0173]	−.1272	+.0072	iny[−.0001,+.0
ng-mR@100	−.0489	−.0724	+.0236	iny[+.0175,+.0296]	−.1549	+.0148	iny[+.0064,+.0

Why. Under the released SUM fusion TDE's change is three subtractions, and the split along them adds up exactly (Appendix O.1): removing the frequency prior moves mean recall by -0.0009; removing the visual term adds $+0.0164$, case-level $+0.0016$, $[-0.0006, +0.0037]$; subtracting the averaged context supplies $+0.0817$ of the 0.0972. That term is, to 99.96% of its energy, one vector shared by every relation in every image, correlated $+0.75$ with the log training prior: in PredCls TDE is to first order a learned logit adjustment of the context branch, which is why the explicit one reproduces it.

Two more methods. DLFE [6] divides predicate probabilities by estimated label frequencies, a per-predicate log-odds shift, and is released with the baseline it was trained against, so this comparison is controlled (Appendix Q.5). Its $mR@50$ gain, $+0.1112$, is $+0.1046$ group-level and $+0.0065$, $[+0.0010, +0.0120]$, case-level; the relation-weighted AUC moves by $+0.0029$, $[-0.0004, +0.0056]$. IETrans [51] relabels training data from general to specific predicates within an object pair and adds the training log-prior at test time; its released Neural-Motifs PredCls checkpoint is the IETrans+Rwt variant, also trained with a reweighted loss, re-scored the same way (Appendix O.2). It reaches $mR@50 = 0.3587$: $+0.2204$ over the baseline, $+0.2143$ group-level and $+0.0060$, $[-0.0031, +0.0151]$, case-level. Its relation-weighted AUC falls by -0.0235, $[-0.0275, -0.0182]$ (by comparison -0.0093, $[-0.0181, +0.0004]$). No IETrans baseline was released, so this compares two separately trained models and does not isolate the effect of relabelling.

Beyond PredCls. The release also holds SGCls and SGDet checkpoints. One TDE pass yields all three models with shared object predictions, so the ground-truth pair remains a common grouping; a replayed evaluator reproduces the released recalls (Appendix Q.6). In SGCls TDE

raises $mR@50$ by $+0.0517$, 99% of it group-level; the control gains $+0.0518$ at $R@50 = 0.384$ against TDE's 0.263; and TDE's within-cell ranking gets worse, by -0.0128 of relation-weighted AUC, interval $[-0.0171, -0.0074]$ (-0.0047, $[-0.0122, +0.0031]$, by comparison; 58% of relations have a matched pair). In SGDet, on the full test set, the control again gains more than TDE ($+0.0418$ against $+0.0310$) at nearly twice its $R@50$.

6. The model the aggregates rank last

The predictors of Sec. 4 never see an image: they read the class pair and, at most, box coordinates from the annotation. On the same reconstructed split we add one that reads pixels — subject and object regions cropped from the image and encoded by a frozen CLIP ViT-B/32 [32], concatenated with the class embeddings MLP-CLASS uses. CLIP is not fine-tuned, so any case-level gain reflects what pretraining had already recovered. All three compared models are retrained on one fixed subsample of $100{,}000$ training relations, so differences isolate the features rather than the amount of data.

By every aggregate measure the CLIP model is the worst of the three: top-1 accuracy 0.6156 against 0.6467 for the geometry model, and a quadratic score 0.04642 lower. An evaluator reading either number would discard it. The split disagrees. Matched under the primary protocol, the CLIP model *gains* 0.00331 of case-level score against the geometry model, interval $[+0.00185, +0.00477]$ ($+0.00464$, $[+0.00311, +0.00618]$, by temperature alone; on raw outputs 0.00640, $[0.00513, 0.00767]$), while losing 0.04705 of group-level score under temperature matching. Its whole deficit is group-level (Fig. 2). Against the class-only model, pixels buy 0.01663 of case-level score on raw outputs where box geometry buys 0.01023, with well-separated intervals.

A larger case-level part is not the same as better ranking, and here the two disagree. With each model's prior matched within cells, the CLIP model still ranks the correct predicate first less often (top-1 accuracy 2.17 points lower, MRR 1.67 lower), and under the primary weighting the within-cell AUC favours geometry, -0.0067, $[-0.0120, -0.0019]$; by comparison the sign reverses, $+0.0085$, $[-0.0062, +0.0214]$. Over five training seeds (Appendix P) the CLIP model's temperature-matched advantage stays between $+0.00413$ and $+0.00474$, its accuracy below geometry's. (It also reads a wider input: two pipelines, not encoders alone.)

Under a label shift. A label shift moves group-level parts and only reweights case-level ones (Sec. 3). Along a path from the benchmark's predicate mix to uniform frequencies within each pair (Appendix Q.3), seven of the eight group-level parts of Sec. 5 and this section change sign and the CLIP model beats geometry by $+0.02981$ at the uniform end. Five of the six case-level parts that excluded zero keep their sign and grow; the exceptions are TDE's, which rises against the baseline from -0.00006 to $+0.01430$, $[+0.01116, +0.01744]$, and against the control from -0.00398 to $+0.00723$, while the AUC separates it from the baseline at neither end: TDE gains discrimination between some predicate pairs and loses it between others, and the benchmark's mix weights the two to zero.

Beyond scene graphs. The supplement applies the split to CIFAR-100-LT, long-tailed text and Waterbirds (cells: background scenes); there the fine class label cannot serve as a cell, since it forces the case-level part to zero.

7. Limitations and conclusion

Limitations. Six bound what the split can say. *It is conditional on the grouping*: two equally fine partitions of the same cases can disagree about whether a gain is case-level (Appendix K), so declaring ϕ is a modelling decision the benchmark should make before a leaderboard opens: refining the class pair by position or size leaves the share at 90%, dropping the object class takes it to 18%. *It credits any within-cell signal*, shortcuts included: an unrecorded covariate correlated at 0.061 with both parts would account for the entire case-level gain of Sec. 4 (Appendix I). *It is not invariant to recalibration, the operating point or the label mix*, so we report two matching protocols; the within-cell AUC, invariant to the first two, gives up exact additivity. *Its interval assumes cells that are not tiny*: with every cell holding two cases it covers 81.5%, from three up it is nominal (Appendix J), and 1.0% of audited relations sit in two-case cells. And *it needs scores for every label*, which most benchmarks do not archive; recovering them took a rerun of the released checkpoint. *Each released model is one training run*: the intervals cover test sampling, not training variability, and the matching is fitted on half of the test images, as no validation outputs were released.

Conclusion. A ten-point mean-recall gain is two achievements summed, and mean recall reports the sum; scoring each prediction against another case's label from the same group separates them exactly. On three released Neural-Motifs checkpoints on Visual Genome the gains are almost entirely group-level: TDE's is matched by a ranking-preserving logit adjustment and its within-group ranking moves by less than 0.0034 under the split's weighting, DLFE's does not move detectably, IETrans's (against a separately trained baseline) falls; we audit no later method. We suggest reporting, beside the headline metric, the declared grouping, both parts with intervals, the within-group AUC under both weightings and, for a thresholded metric, the operating-point control; archived top-k logits suffice for the AUC and the control, not for the exact split.

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